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Luddy 25 – Research symposium highlights Luddy School's 'transformative' work

Transformative research at what Luddy Dean Joanna Millunchick calls the “intersection of theory and real-world impact” was highlighted during Luddy's 25th anniversary research symposium.

Feb 24, 2026



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A junior faculty panel discussion was among the symposium highlights.

Photo by Aaron Smith, Luddy School communications.

The [Luddy School of Informatics, Computing, and Engineering's](#) difference-making research in areas ranging from computer vision to AI to biological science to robotics and so much more was showcased during last week's [Luddy School 25th Anniversary Celebration of Research](#) at Luddy Hall.

Held in conjunction with the school's 25th anniversary, Luddy Dean [Joanna Millunchick](#) said the symposium demonstrated how the school is “fulfilling [Fred Luddy's](#) vision of seeing tomorrow while others are still figuring out today.”

“As we look at the last quarter century,” she said, “we see a trail of firsts. It is clear the next 25 years will be even more transformative. Research is our lifeblood.”



David Crandall, Luddy professor of computer science, spoke on computer vision and its promising research possibilities.

Credit: Photo by Chris Kowalczyk, Luddy School communications.

Transformation came in many forms during the day-long event that drew more than 100. [David Crandall](#), Luddy professor of computer science and director of the Luddy Artificial Intelligence Center delivered an entertaining presentation centered on computer vision – “the goal is to study how to make computers perceive the world around them,” he said – progress enlightenment and his own “mid-life crisis.”

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That included the realization that, 25 years after attending his first Computer Vision and Pattern Recognition Conference at a Wyoming bison ranch (he had the photo to prove it), Crandall could say the computer vision community has “grown dramatically” even as many computer vision papers aren’t “really studying the problems in the vision community that matter to real people.”

Crandall’s research drives to change that. One example – he said young children are “amazing visual learners” and that the “visual experiences they get seem to be very good training not only for them, but for computers.”

Crandall’s work puts that to cutting-edge use.

[Funda Ergun](#), Luddy professor of computer science and associate dean for research, said the symposium’s goal was to highlight research at Luddy at all levels. The event featured a plenary talk and “high-level discussions from faculty of every seniority and department.” A panel highlighted junior faculty research, while a poster session recognized graduate student research.

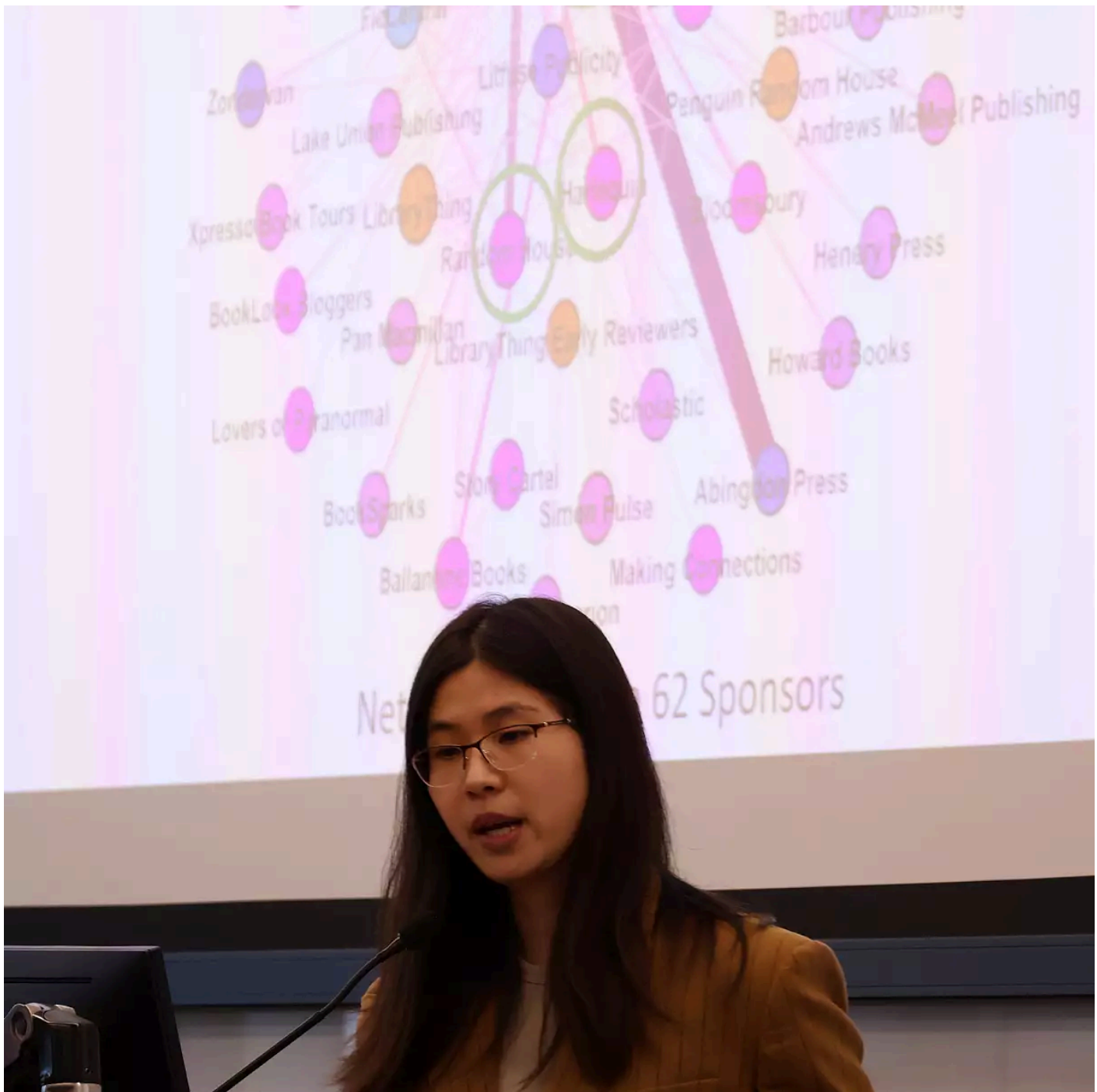
Ergun called it a “celebration of working together on a variety of topics as the sessions (were) grouped according to general approach to science and goals, such as theory vs. practice, real-life research, and research for humanity.”

Ergun added it also was a celebration of “our unity as a school.”

Ishwari Dawkhar, a second-year data science master’s student, described the symposium, specifically the plenary talk by [Benjamin Pierce](#), Henry Salvatori Professor of computer science at the University of Pennsylvania, as “very refreshing.”

“It’s given me a lot of good points, especially about AI coding,” she said. “The professor was great about explaining the fundamentals of why things do and don’t work, and how we can get better at these systems.”
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Dawkhar added that such information is helpful in bridging the gap between academic requirements and industry needs.



Yuerong Hu, assistant professor of information and library science, demonstrated critical cultural analytics through book reviews.

Credit: Photo by Aaron Smith, Luddy School communications.

Millunchick called the founding of the Luddy School 25 years ago a “bold experiment” that has “grown into a world-class engine of innovation.”

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Luddy School Dean Joanna Millunchick emphasized the school's "transformative research."

Credit: Photo by Chris Kowalczyk, Luddy School communications.

Then known as the IU School of Informatics, it was the first of its kind in the United States. Millunchick said the approach that "technology should not be studied in isolation but through the lens of how it impacts people, society and every other field of study" was "radical" at that time.

Millunchick said the research symposium's schedule was structured to "show how our research moves from the chalkboard to the community."

The first session addressed how Luddy researchers solve complex problems in vision and biological sciences. The second session focused on research on robotics and innovations in medical imaging and non-invasive diagnostics.

"Quite literally," Millunchick said, "we're building the future."

The third session discussed the human-centric side of technology.

"How do we visualize data?" Millunchick said. "How do we interact with social robots and ensure that the intelligence we build remains centered on human needs?"

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Highlights were many.

Pierce, an [ACM Fellow](#) who has won the award for the most influential [Principles of Programming Languages](#) paper, worked at Indiana University in the late 1990s before moving on to the University of Pennsylvania.



University of Pennsylvania professor Benjamin Pierce, who once worked at IU, gave the plenary talk.
Credit: Photo by Chris Kowalczyk, Luddy School communications.

During his *Properties for the People* talk, he discussed recent advances in Property-Based Testing, which is a lightweight formal method offering an attractive blend of the abstract (formal specification) and the concrete (executability). He said its popularity in the functional programming community has grown steadily over the past 25 years, and it is poised to play a key role in the larger software industry in mediating between human programmers and their AI collaborators.

[Jeremy Siek](#), professor of computer science, said Pierce's early work, much of it from his IU days, is having major impact today, and that his programming language textbook became the "the bible of how we do research."

"That's emblematic of Benjamin's work, so much of it helps the community," Siek said, adding that the "resulting technology has blossomed. We use it day in and day out. Mathematicians are using it. Giant projects are verifying some pretty amazing theorems by using this technology."



A poster session highlighted some of the students' work.

Credit: Photo by Chris Kowalczyk, Luddy School communications.

[Lantao Liu](#), associate professor of computer science and director of the Vehicle Autonomy and Intelligence Lab, discussed the challenges of translating robotics research breakthroughs into reliable real-world systems, Luddy's impressive success in autonomous car racing and the launching of the robotics degree that will further enhance the school's reputation as a "national hub for robotics and autonomy."

[Eleftherios Garyfallidis](#), associate professor of intelligent systems engineering, addressed "Minimalist AI." He said while large AI systems have become more powerful and we are also using or developing such systems, they are still hard to control and understand. At the opposite of those systems are smaller systems. He wondered if smaller systems can be more effective than larger ones.

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He introduced an advanced reconstruction model called FORCE for FORward modeling for Complex microstructure Estimation. It's a new way to analyze not just

medical MRI scan data, but others, as well.

The first research session, Theory Meets Practice, consisted of Crandall; [Qin Zhang](#), adjunct professor in mathematics and professor of computer science; [Paul Macklin](#), associate professor of intelligent systems engineering and associate dean for undergraduate education; [Mohsen Heidari](#), assistant professor of computer science; and [Arjan Durresi](#), professor of computer science (IU Indianapolis).

Zhang discussed collaborative learning with limited communication. Macklin talked about integrating human and machine intelligence in cancer simulations. Heidari focused on quantum machine learning beyond classical simulatability. Durresi's topic was casualty and world models as foundations for AI control.

The second session, Research Meets Applications, consisted of Liu; Garyfallidis; [Rakesh Shiradkar](#), assistant professor biomedical engineering and informatics (IU Indianapolis); [Cici Ling](#), assistant professor; [Feng Guo](#), associate professor of intelligent systems engineering; and [Elizabeth Kaziunas](#), associate professor of informatics.

Shiradkar addressed integrating multi-modal imaging with AI for improved precision medicine; Ling targeted understanding abusive behavior on social media through a mixed-method and cross-platform multimodal approach. Guo discussed organoid meets AI and medicine.

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Katy Börner, Victor H. Yngve professor of engineering and information science, addressed visualizing multiscale complex systems.

Credit: Photo by Aaron Smith, Luddy School communications.

The third session, Machine Meets Human, was conducted by [Samantha Wood](#), assistant professor of informatics; [Katy Börner](#), Victor H. Yngve professor of engineering and information science; [Staća Milojević](#), professor of informatics; [Noriko Hara](#), professor of information science; [Rongqian Ma](#), assistant professor of information and library science; and [Yuerong Hu](#), assistant professor of information and library science.

Wood discussed reverse engineering the origins of intelligence. Börner addressed visualizing multiscale complex systems. Milojević talked about mapping the unpredictable and understanding the social dynamics of human innovation. Hara focused on GenAI and lay contribution to scientific knowledge on social media. Ma's presentation was, "A Dancing Bear, A Colleague, or A Sharpened Toolbox? The Cautious Adoption of Generative AI Tools in Digital Humanities Research." Hu [Skip to main content](#) provided critical cultural analytics through book reviews.



Samantha Wood, assistant professor of informatics, discussed reverse engineering the origins of intelligence. Credit: Photo by Aaron Smith, Luddy School communications.

Through it all, Millunchick said, the Luddy School has “evolved from a small single department to a multi-campus powerhouse that spans from the small (like bioinformatics) to the large (global networks and AI-driven research). Today, the program we have reflects that Luddy DNA. It’s the intersection of theory and real-world impact.”

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Indiana University

Pete DiPrimio

Writer/Editor

P: [\(812\) 855-1341](tel:(812)855-1341)

E: pdiprimi@iu.edu

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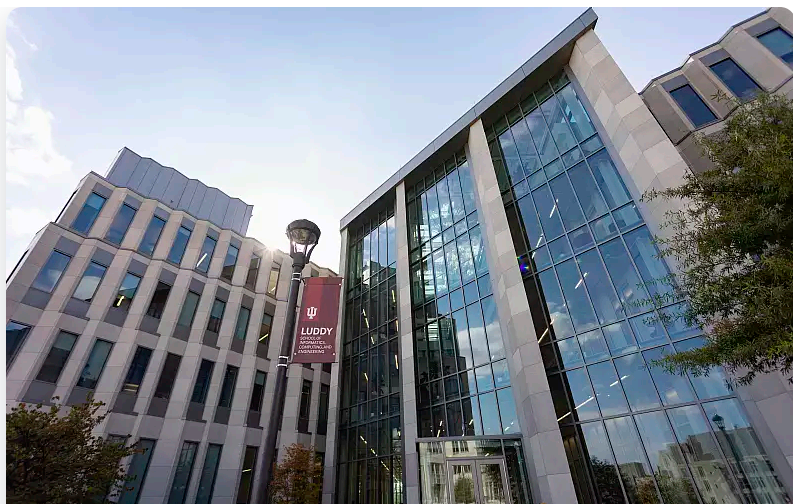
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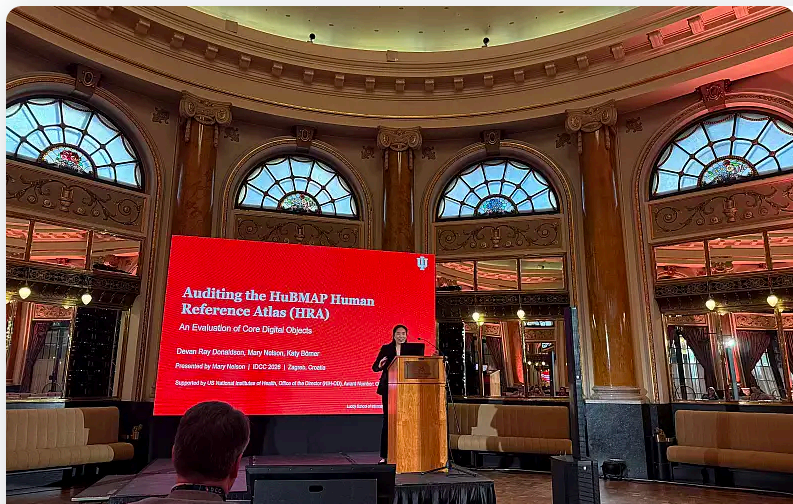
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Luddy Bloomington

700 N. Woodlawn Avenue

Bloomington, IN 47408

812-856-5754

luddy@iu.edu

Luddy Indianapolis

535 W. Michigan Street

IT 400, Indianapolis, IN

317-278-4636

luddyin@iu.edu

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